

Lecture 5: The System Bus & Instruction Cycle (MCQs)

1. A _____ is a shared pathway consisting of a collection of wires connecting CPU, Memory, and I/O.

- a) Register b) System Bus c) Control Unit d) Cache

• **Answer:** b) System Bus

2. In the bus model, the device that initiates the data transfer is called the _____.

- a) Slave b) Master c) Arbiter d) Decoder

• **Answer:** b) Master

3. Which type of bus timing requires a "Handshaking" protocol instead of a global clock?

- a) Synchronous b) Semi-synchronous c) Asynchronous d) Dedicated

• **Answer:** c) Asynchronous

4. In a Synchronous bus, master outputs are valid on the _____ edge of the clock (CLK).

- a) Falling b) Rising c) Middle d) End

• **Answer:** b) Rising

5. Which bus setup is best suited for applications where messages are generated at irregular intervals (like keyboards)?

- a) Synchronous b) Asynchronous c) Semi-synchronous d) Multiplexed

• **Answer:** b) Asynchronous, (*Exam 2025 Question*)

6. In a Semi-synchronous bus, extra clock cycles inserted due to a late ACK are called _____.

- a) Wait states b) Deadlocks c) Interrupts d) Strokes

• **Answer:** a) Wait states

7. The _____ Bus is bi-directional, allowing data to flow in both directions.

- a) Address b) Control c) Data d) Power

• **Answer:** c) Data, (*Exam 2025 Question*)

8. The width of the _____ Bus determines the maximum memory addressing capacity.

- a) Data b) Address c) Control d) System

• **Answer:** b) Address

9. If a computer has a 32-bit Data Bus, it means it has _____.

a) 32 memory locations b) 32 wires to carry data simultaneously c) 32 I/O ports d) 32 instructions

• **Answer:** b) 32 wires to carry data simultaneously

10. A bus that uses the *same* lines for both Address and Data at different times is called _____.

a) Dedicated b) Parallel c) Multiplexed d) Serial

• **Answer:** c) Multiplexed

11. What is the main disadvantage of a Multiplexed bus?

a) Requires more pins
b) More expensive
c) Complex control circuitry and slower performance
d) Cannot handle interrupts

• **Answer:** c) Complex control circuitry and slower performance

12. The ISA bus was originally introduced by IBM as an _____ bus.

a) 8-bit b) 16-bit c) 32-bit d) 64-bit

• **Answer:** a) 8-bit

13. PCI Bus typically uses _____ lines for address and data.

a) Dedicated b) Multiplexed c) Separate d) Asynchronous

• **Answer:** b) Multiplexed

14. Which bus standard is known for being "Hot Pluggable" and uses a serial connection?

a) ISA b) PCI c) PCI Express (PCIe) d) AGP

• **Answer:** c) PCI Express (PCIe)

15. A PCIe connector is typically intended for use with _____.

a) Network cards b) Sound cards c) Graphics ports (Video cards) d) Keyboards

• **Answer:** c) Graphics ports (Video cards)

16. USB 1.1 (Full-speed) supports a transfer rate of _____.

a) 1.5 Mbps b) 12 Mbps c) 480 Mbps d) 1 Gbps

• **Answer:** b) 12 Mbps

17. USB 2.0 (Hi-speed) is capable of supporting a transfer rate of up to _____.

a) 12 Mbps b) 100 Mbps c) 480 Mbps d) 5 Gbps

• **Answer:** c) 480 Mbps, (Exam 2025 Question)

18. The maximum number of devices supported by USB is _____.

a) 63 b) 64 c) 127 d) 255

• **Answer:** c) 127

19. Which control signal indicates that data on the bus is to be written into the addressed location?

a) Memory Read b) Memory Write c) Bus Grant d) Transfer ACK

• **Answer:** b) Memory Write

20. The signal used to initialize all modules is called _____.

a) Clock b) Interrupt c) Reset d) Halt

• **Answer:** c) Reset

21. In the Asynchronous Handshaking protocol, the sends a _____ signal.

a) ACK b) STROBE c) WAIT d) GRANT

• **Answer:** b) STROBE

22. In the Asynchronous Handshaking protocol, the responds with a _____ signal.

a) ACK b) STROBE c) RESET d) CLOCK

• **Answer:** a) ACK

23. During the phase, the first operation is typically: _____.

a) $IR \leftarrow MBR$ b) $MAR \leftarrow PC$ c) $PC \leftarrow PC + 1$ d) $MBR \leftarrow Memory$

• **Answer:** b) $MAR \leftarrow PC$

24. The register that holds the data fetched from memory before it goes to the IR is _____.

a) MAR b) PC c) MBR d) ALU

• **Answer:** c) MBR

25. Which phase of the instruction cycle analyzes the instruction to determine the operation?

a) Fetch b) Decode c) Execute d) Store

• **Answer:** b) Decode

26. Which bus architecture is "cheaper" because it requires less hardware?

a) Synchronous b) Asynchronous c) Dedicated d) Parallel

• **Answer:** b) Asynchronous

27. A major advantage of the Synchronous bus is _____.

- a) Flexibility for different speeds
- b) Lower overhead and greater throughput
- c) No clock required
- d) Handshaking protocol

• **Answer:** b) Lower overhead and greater throughput

28. The PCI Express connector has _____ pins.

- a) 36
- b) 64
- c) 98
- d) 164

• **Answer:** a) 36

29. Which register is incremented to calculate the address of the next instruction?

- a) IR
- b) MAR
- c) PC
- d) MBR

• **Answer:** c) PC

30. Control Bus carries signals for _____.

- a) Data only
- b) Addresses only
- c) Control and co-ordination of activities
- d) Power supply only

• **Answer:** c) Control and co-ordination of activities, (*Exam 2025 Question*)